



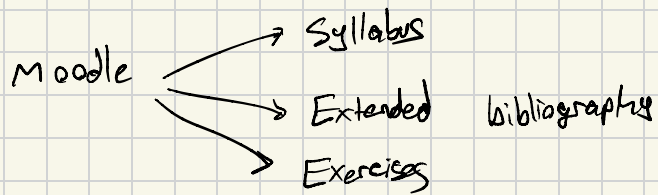
Andreas Goudelis

Indicative bibliography

- M. Blau "Lecture notes on General Relativity"
→ mostly chapters: 1-3, 8, 19, 23, 29, 33, 34
- B. Schutz "A first course in GR"
- Hobson, Efstathiou, Lasenby: "GR: An introduction for physicists"

Exam

→ No electronic devices



1) Introduction

- 4 fundamental interactions

- i) EM: Maxwell's eqs / QED
 - ii) Weak interaction:
 - iii) Strong: QCD
 - iv) Gravity: Newtonian / general relativity
- } → The SM
 $SU(2)_L \times U(1)_Y$
electroweak theory

• Why GR?

i) it is the 4th interaction

ii) It is not quantum yet

iii) Particle physics in the cosmos $\rightarrow$ shortcomings of the SM

Dark matter $\rightarrow$ Baryon asymmetry of the universe

GR $\rightarrow$ A theory of gravity
A theory of space-time $\rightarrow$ J. A. Wheeler: S-T tells matter how to move and matter tells S-T how to curve

2) Useful notions

• Event: A physical occurrence in space and time (associated to a specific S-T point)

Remark: An event has no spatial extension and no duration.

Ex: Emission of a photon

— The presence of a point-like particle at some S-T point.

• Observers: Essentially, a reference frame

Note: We can switch reference frames through coordinate transformations

• Local time: Each observer has a notion of time and all observers measure time in the same way

• Worldline: The path that a point-like object follows in Space and Time

Note: For extended objects: World sheets and World volumes

3) The equivalence principle (E.P.)

A) Newtonian / Galilean version

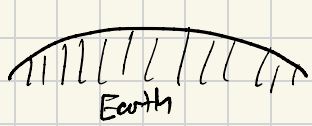
M_g $\bullet$ P $\downarrow \vec{F}_g$

- From Newtonian gravity $\rightarrow \vec{F} = -M_g \vec{\nabla} \phi$ M_g : "gravitational mass"

- From Newtonian mech $\rightarrow \vec{F} = M_i \vec{a}$ M_i : "inertial mass"

$M_i \stackrel{?}{=} M_g$

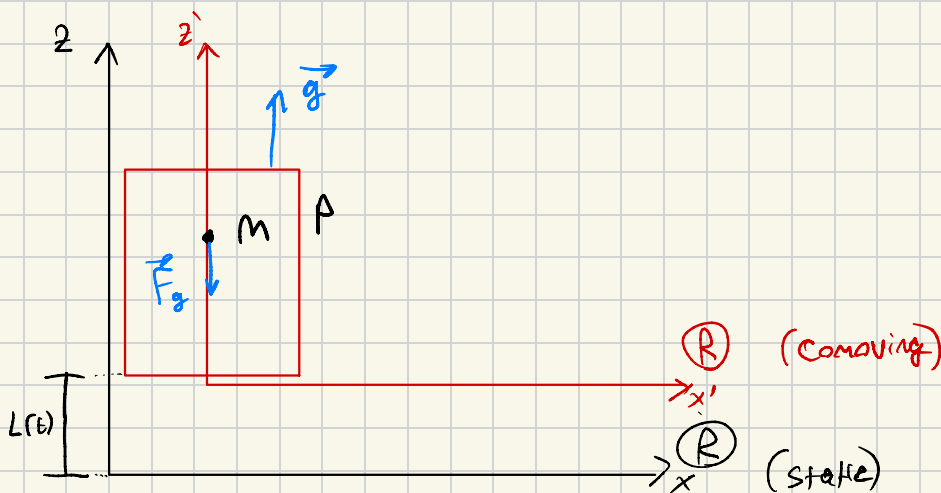
Yes, because we observe so



Galilean equivalence principle: $M_i = M_g$

$$\Rightarrow \vec{a} = -\vec{\nabla} \phi \iff \text{universality of free fall}$$

B) Einstein's version of the E.P.



- Case 0: Elevator in outer space, no forces $\Rightarrow$ P just floats
- Case 1: Still in outer space, elevator accelerated, w/const acc $\vec{g}$
 $\Rightarrow$ P approaches the floor
- Case 2: Elevator brought in const. gravitational field
 $\Rightarrow$ P approaches the floor

$\rightarrow$ Both R and R' agree on these statements

But R' cannot distinguish 1 from 2

$\rightarrow$ Gravity = Acceleration

$$L(t) = \frac{1}{2} g t^2 \Rightarrow \ddot{L}(t) = g \quad \text{and} \quad z' = z - \frac{1}{2} g t^2$$

$$t' = t \quad (\text{ignoring SR})$$

- In R : $F = m \ddot{z}$

- In R' : $F' = m \ddot{z}' \Rightarrow F' = m \ddot{z} - mg \Rightarrow F' = F - mg$

$\Rightarrow$ the acceleration of R' wrt R looks exactly like gravity in R'

• Case 3 : Still in the gravitational field as we cut the elevator rope

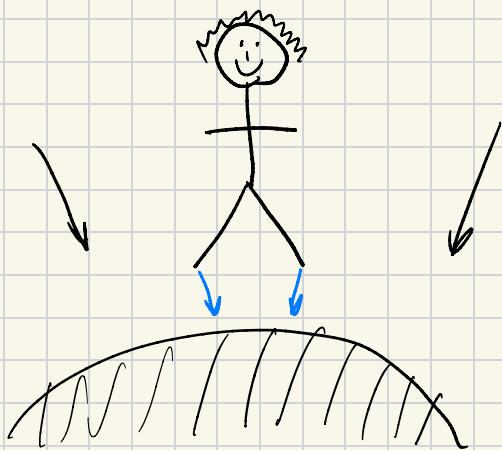
$\Rightarrow$ P floats for R'

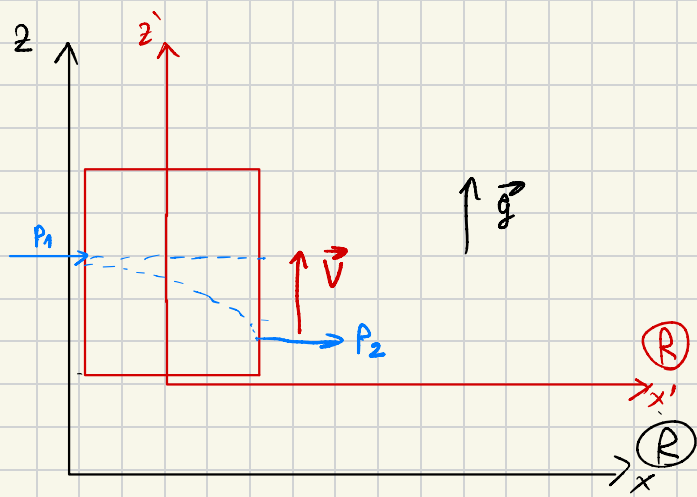
$\Rightarrow R'$ cannot tell if he/she is static in outer space or freely falling

The equivalence Principle : No physics experiment can distinguish locally between a uniform grav. field acting on the local inertial observer and a uniform acc.

wrt that local inertial observer.

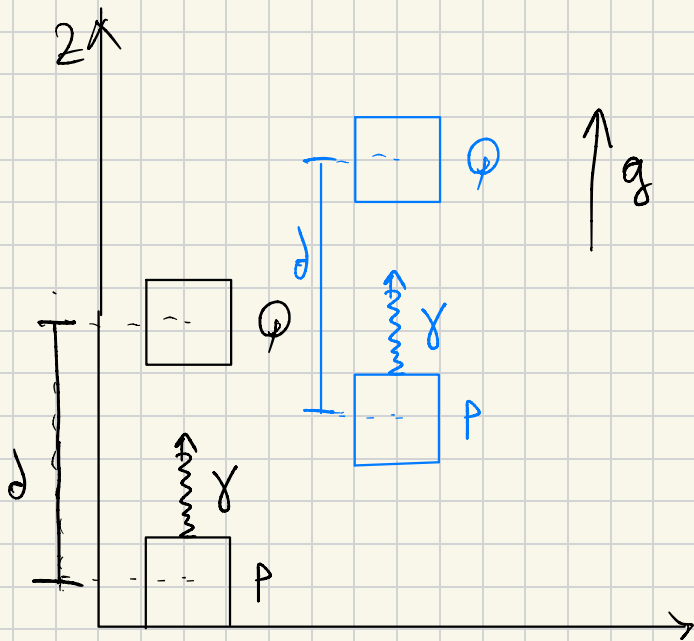
$\hookrightarrow$ The one for whom SR holds





Exercise

- Gravitational redshift



- photon of frequency f_p emitted from A

Φ : frequency measured by $\Phi(F_\theta)_\theta$?

Sol⁹ Photon ToF $\Delta t = d/c$

P: emitter
Q: Receiver

During the same time the velocity of PQ (S) increases by $\Delta v = g \Delta t = \frac{g d}{c}$

In the acc. R.F. $\odot$ will see a doppler-shifted frequency f_D

$$\frac{F_0}{F_P} = 1 - \frac{\Delta V}{c} = 1 - \frac{g d}{c^2}$$

Equivalence Principle: Acceleration $\Leftrightarrow$ gravity

$\Rightarrow$ In the presence of a gravitational field, and in a fixed RF, photons will be redshifted as $\frac{f_Q}{f_P} = 1 - \frac{gd}{c^2}$

Equivalently: clocks at Q tick slower than clocks at P.